Reg. No. | | | | | | | | | | |

[13AE55]

FIFTH SEMESTER B.E EXAMINATION, DECEMBER–2015

HEAT AND MASS TRANSFER

Time:3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions.
2. Draw neat sketches wherever necessary.
3. Assume suitably any missing data & state it.

1. a. Explain the three mechanisms of heat transfer with their respective equations for heat transfer rates. **06**

b. A long homogenous resistance wire of radius R_o=0.6cm and thermal conductivity k=15.2 w/m.K is being used to boil water at atmospheric pressure by the passage of electric current. Heat is generated in the wire uniformly as a result of resistance heating at a rate of 16.5 W/Cm^3. The heat generated is transferred to water at 100°C by convection with an average heat transfer coefficient of h=3200w/m^2.K. Assuming steady one-dimensional heat transfer: **08**

i. Express the differential equation and boundary conditions for heat conduction through the wire.

ii. Obtain a relation for the variation for temperature in the wire by solving the differential equation.

iii. Determine the temperature at the centreline of the wire

2. A square plate heater of $15\times15\,cm^2$ area is inserted between two slabs. Slab A is 2cm thick (k=500w/m.K) and slab B is less thick (k=0.2w/ m.K). The outside heat transfer coefficients on side A and side B are 200w/m^2.K and 50W/m^2.K respectively. The temperature of surrounding air is 25°C. If the rating of the heater is 1KW, find:

i. The maximum temperature in the system **06**

ii. Outer temperature of the two slabs **06**

iii. Draw an equivalent electrical circuit **02**

3. a. Define fin efficiency and develop the mathematical relation for fin efficiency for fins with adiabatic tip. **04**

b. Aluminum fins of rectangular profile are attached to a plane wall with 5mm spacing. The fins have thickness of 1mm, length 10mm and thermal conductivity of 200W/m, K and the fins dissipate heat by convection into an ambient at 40°C **10**

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with a surface heat transfer coefficient of 50W/m^2.K. The wall is maintained at 200°C. Determine

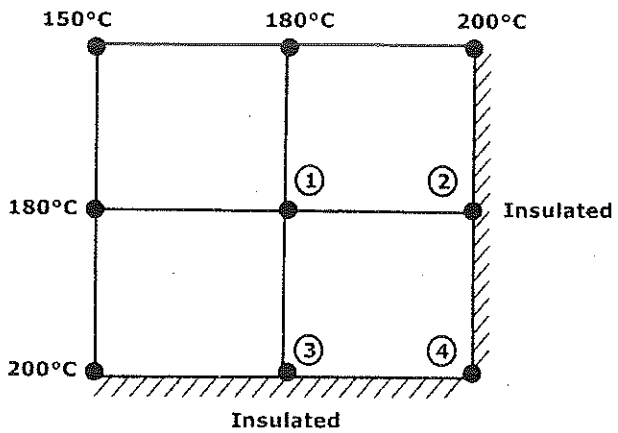
i. The fin efficiency

ii. The heat loss from the plane wall per m^2 of the wall surface.

Neglect the heat loss from the fin tip.

4 a. Explain in brief, lumped system analysis in transient heat conduction. Discuss the condition under which the above analysis is valid. **04**

b. Consider steady two-dimensional heat transfer in a long solid bar of square cross section as shown in figure 1. The measured temperatures at selected points of the outer surfaces are as shown. The thermal conductivity of the body is K=20W/m°C and there is no heat generation. Using the finite difference method with a mesh size of $\Delta X = \Delta Y = 1.0$cm, determine the temperatures at the indicated points in the medium. **10**



5. a. Explain briefly about development of the thermal boundary layer in the entrance region of a circular tube. How does the Prandtle number influence the relative growth of hydrodynamic and thermal boundary layers? The tube is maintained at a constant temperature higher than the approaching fluid. **06**

b. The top surface of the passenger coach of a train moving at a velocity of 70km/h is 2.8m wide and 8m long. The top surface is absorbing solar radiation at a rate of 200W/m^2 and the temperature of the ambient air is 30°C. Assuming the roof of the coach to be perfectly insulated and the radiation heat exchange with the surroundings to be small relation to convection. **08**

i. Write the heat balance equation under steady state condition.

ii. Determine the average heat transfer coefficient

iii. Determine the equilibrium temperature of the top surface of the coach

Properties of air are $\nu = 1.608 \times 10^{-5} m^2/s$, P_r=0.7282, k=0.02588W/m, K

Use the connection $N_u = \left[0.037 R_{el}^{0.8} - 871.3\right] P_r^{1/3}$

6. a. State the following laws of radiation with their mathematical expressions: **06**

i. Stefan boltzmance law

ii. Planck's law

iii. Wien's displacement law

b. Two large parallel plates are at 1000K and 800K. Determine the heat exchange per unit area when: **08**

i. The surfaces are black

ii. The hot surfaces has an emissivity 0.9 and the cold 0.6

iii. A large plate of emissivity 0.1 is inserted between them in case (ii) above

Also find the percentage reduction in heat transfer, because of introduction of the large plate. $\sigma = 5.67 \times 10^{-8} w/m^2, K^4$

7. a. Derive an expression for LMTD for the analysis of counter flow heat exchanger. **07**

b. Steam enters a counter flow heat exchanger at 350°C. The mass flow of steam is 800kg/min. Hot gases enter the heat exchanger at 650°C and mass flow rate is 1350kg/min. If the tubes are 300mm diameter and 3m long, determine the number of tubes required. Neglect the thermal resistance offered by the metallite tubes. **07**

C_P (steam)=3.772kJ/kg °C, C_P (gas)=1.0kJ/kg °C

h (steam) = 600W/m^2 °C, h (gas) = 250W/m^2 °C

8. a. Explain Fiek's law of diffusion for a binary mixture on **04**

i. mass basis

ii. mole basis

b. Explain physical significance of **04**

i. schmidt number

ii. Levis number

c. Pressurized hydrogen is stored at 358K in a 4.8m outer diameter spherical container made of Nickel. The shell of the container is 6cm thick. The molar concentration of hydrogen in the nickel at the inner surface is determined to be 0.087 kmol/m^3. The concentration of hydrogen in the nickel at the outer surface is negligible. Determine the mass flow rate of hydrogen by diffusion through the nickel container. The binary diffusion coefficient for hydrogen in the nickel at the specified temperature is 1.2×10^{-12} m^2/s. **06**

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